

A Simple Approach to Animating Virtual Characters by Facial Expressions Reenactment

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ABSTRACT

Animating virtual characters is one of the core problems in virtual reality. Facial animations are able to intuitively express emotion and attitudes of virtual characters. However, creating facial animations is a non-trivial task. It depends on either expensive motion capture devices or human designers' time and effort to tune the animation parameters. In this work, we propose a learning-based approach to animate virtual characters by facial expression reenactment from abundant image data. This approach is simple yet effective, and is generalizable to various 3D characters. Preliminary evaluation results demonstrate its effectiveness and its potential to accelerate the development of VR applications.

Index Terms: Virtual Human—Facial Animation—Face Reenactment—Blendshape

1 INTRODUCTION

Virtual reality (VR) technology is attracting increasing attention as it provides users with an immersive experience. When building virtual characters, it is of great significance to create vivid facial animations, because they can intuitively convey emotion and feeling, which is crucial in many VR applications [1]. However, it is non-trivial to create diverse facial animations conveniently. Traditional methods usually use motion capture devices to track the key points of human actors and replicate them on virtual characters. The requirement of the dedicated devices limits its application. Recently, most 3D virtual characters are animated by a linear combination of muscle movements under the control of corresponding coefficients. A popular implementation among them is blendshape. In this scheme, creating animations usually requires human designers to tune the coefficients in a trial-and-error manner, which is boring and time-consuming.

In this paper, instead of creating animations from scratch, we propose to address this problem by facial expression reenactment based on abundant image data. Previous methods of face reenactment mainly focus on the 2D image/video domain, while few methods explore its application in 2D to 3D scenario. Under the popular blendshape scheme, we postulate that the generation of facial animation is essentially an estimation of the blendshape coefficients. Therefore, we propose to achieve facial expression reenactment by estimating blendshape coefficients.

The approach is designed to be simple in theory and practice in order to enhance usability. As illustrated in Fig. 1, the approach contains two models. We first pre-train a base model that estimates generic 3D facial parameters from the given image. Based on the frozen base model, we train a lightweight adapter model to adapt the generic parameters into the desired blendshape coefficients of the target virtual character. During testing, the pipeline takes human face

images as input and estimates corresponding blendshape coefficients. This approach is able to reduce the workload of human designers in creating facial animations. Besides, the paradigm of a frozen base model plus a trainable lightweight adapter makes this approach generalizable. On one hand, once trained, it is generalizable to 3D characters with the same blendshape topology, even with different texture appearances. On the other hand, the adapter model is a lightweight framework that can be easily retrained to adapt to a new blendshape topology. Preliminary evaluation results demonstrate its effectiveness and its potential to accelerate the development of VR applications. We will make our solution public.

2 APPROACH

2.1 Model Details

The linear 3D Morphable Model (3DMM) is employed as the generic 3D face representation, in which the face shape $\mathbf{S}$ and texture $\mathbf{T}$ in 3D space is represented as: $\mathbf{S} = \bar{\mathbf{S}} + \alpha\mathbf{B}_{id} + \beta\mathbf{B}_{exp}$, $\mathbf{T} = \bar{\mathbf{T}} + \sigma\mathbf{B}_{tex}$, where $\bar{\mathbf{S}}$ and $\bar{\mathbf{T}}$ are mean face shape and texture respectively. $\mathbf{B}_{id}$, $\mathbf{B}_{exp}$, and $\mathbf{B}_{tex}$ denote PCA bases of identity, expression, and texture respectively. In addition, we also introduce illumination model and camera model to define the light coefficient γ and pose coefficient p respectively. All the coefficients mentioned above are concatenated into a single vector $\mathbf{v} = (\alpha, \beta, \sigma, \gamma, p)$. The base model is implemented as a ResNet-50 with a modified projection head to fit the output dimension. The model takes human face images as input and predicts the 3DMM coefficients vectors.

For facial expression reenactment, the adapter model takes the expression coefficient β and pose coefficient p as input and predicts blendshape coefficients $\hat{\gamma}$. The adapter is implemented as a lightweight Multi-Layer Perceptron (MLP) neural network. Specifically, after the last layer, there is a *Clamp* operator truncating the output value into the range $0 \sim 1$, which enforces the model to output valid values of coefficients.

Generally, the pipeline can be formulated as:

$$\alpha, \beta, \sigma, \gamma, p = \text{BaseModel}(x), \quad (1)$$

$$\hat{\gamma} = \text{Adapter}(\beta, p), \quad (2)$$

where x denotes the input face image.

2.2 Training Strategy

The two models are trained in consecutive order. Once trained, the base model serves as a foundation and the training of the adapter model relies on the pre-trained base model.

We use a public real-world face dataset to train the base model in a self-supervised way. The ResNet backbone estimates the 3DMM parameter vector $\hat{\mathbf{v}}$, which is used to reconstruct the 3D face and render it back to the image plane $\hat{x}$ with the differentiable renderer. The model is optimized by minimizing the reconstruction error between x and $\hat{x}$. In addition, we also employ perception loss and landmark loss introduced in 3D face reconstruction work [2].

To train the adapter model, we prepare a dedicated dataset for the target 3D character. The dataset consists of: 1) randomly generated blendshape coefficients; 2) correspondingly rendered images of the

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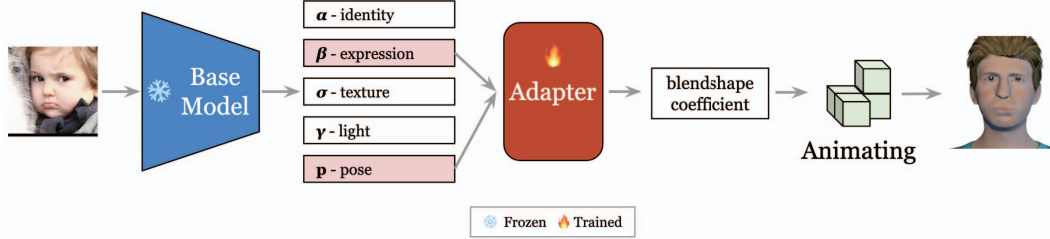


Figure 1: The base model estimates generic 3D facial parameters. Once trained, it is set to frozen state. The adapter model estimates blendshape coefficients based on expression and pose coefficients. The adapter is a lightweight model that can be adapted to various characters.

virtual character’s face. Note that the dataset preparation is an automatic process with the help of 3D software, like Maya. During preparing blendshape coefficients, a filtering mechanism is designed to get rid of contradictory facial blendshapes. For example, a person can hardly move his lip in two opposite directions simultaneously. Empirical studies show that a set of manually designed rules is powerful enough to filter a large portion of contradictory samples. The adapter model is trained on this dataset by minimizing the Mean Squared Error (MSE) between ground-truth blendshape coefficients y and estimated blendshape coefficients $\hat{y}$.

2.3 Usage

Given the pre-trained models, the way of creating animations depends on the type of source data. If the source data is a still image, users can estimate blendshape coefficients of the given image, generate facial expression as the peak intensity, and create facial animation by a smooth transition from natural expression to peak intensity expression. If the source data is a video, users can estimate blendshape coefficients of each frame and then generate animation easily.

The two models in this pipeline play different roles. The base model aims to provide generic 3D facial parameters, which is a character-agnostic model, i.e., training and using of the base model are not concerned with the 3D virtual character. In contrast, the adapter model is a character-dependent model. Training and using of the model are coupled with specific blendshape topology.

The generalization of the approach is reflected in two folds. First, once trained, it generalizes well in the 3D character family with the same blendshape topology, even with different texture appearances. This has covered a large portion of the usage scenario. Second, when adapting the approach to a new blendshape topology, users only need to retrain the adapter model with the automatically generated dataset regarding the target 3D character. In addition, thanks to the simple and light design of the adapter model, the retraining requires little computing resources and takes little time, enhancing its usability.

3 EVALUATION

3.1 Qualitative Examples

In Fig. 2, we show examples with various virtual character gender and appearance. The reenacted facial expressions are capable of vividly replicating the source facial expressions, validating the effectiveness of our approach. Even though there are some subtle muscle movements that are not reproduced perfectly, the estimated coefficients still provide us with a fair reasonable initial state.

3.2 Comparison with human designers

We compare the proposed approach with human designers’ performance in terms of satisfaction score and time usage. The satisfaction score, ranging from 1 to 10, is obtained from 14 volunteers scoring on 7 randomly picked samples. The result is shown in Tab. 1. Though the human designer achieves a higher satisfaction score than the proposed approach, the margin is limited. Besides, the variation of the score is relatively large according to the standard deviation, which reflects the subjectiveness of measuring facial expressions. For the inference time to generate or adjust blendshape coefficients,



Figure 2: Examples of virtual characters with different gender and appearance.

the proposed approach costs only 0.41s for running the whole testing pipeline, while the human designer need over 236s to tune the blendshape coefficient values into the desired state. The huge gap shows the effectiveness of our approach to reducing designers’ workload without losing much fidelity.

Table 1: Comparison with human designers.

	Satisfaction Score	Inference Time
Algorithm	6.36 ± 1.80	$0.41 \pm 0.06s$
Designer	6.92 ± 1.91	$236.12 \pm 48.61s$

4 CONCLUSION

In this work, we propose a simple yet effective approach to ease the burden of creating facial animations of virtual characters for VR applications. The task is formulated as a facial expressions reenactment problem and is achieved by estimating blendshape coefficients based on image data. This approach not only generalizes well within the character family with the same blendshape topology but is also easy to be adapted to other customized characters. Evaluation results have shown promising performance on this task and verified the effectiveness of our approach. We hope this work will inspire research on automatically animating virtual characters.

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